

§ 12.4 Differentiability.

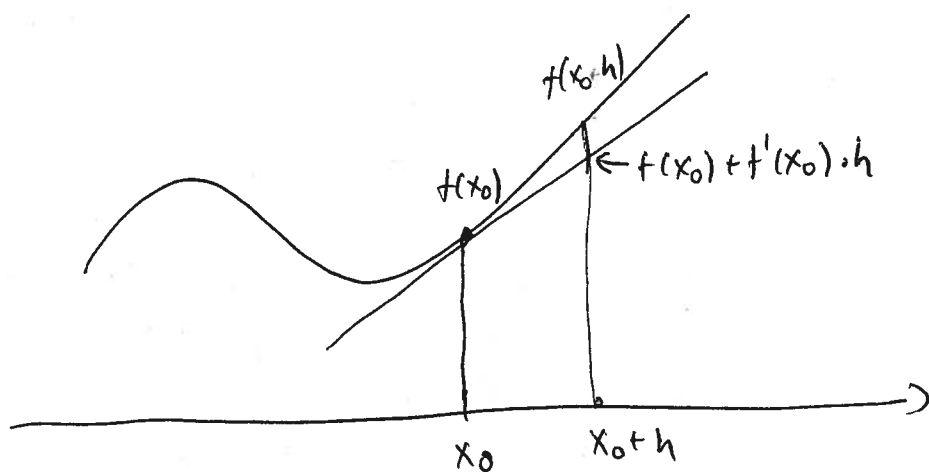
In one variable $f = f(x)$

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\epsilon \rightarrow 0} \frac{f(x_0 + \epsilon) - f(x_0)}{\epsilon} \\ = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

We have $f(x_0 + h) = f(x_0) + f'(x_0)h + h \cdot \epsilon(h)$

where $\lim_{h \rightarrow 0} \epsilon(h) = 0$

equivalently $\lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} - f'(x_0) \right] = 0$



Defn $f(x, y)$ is differentiable at (x_0, y_0) if

$$f(x_0 + h, y_0 + k) = f(x_0, y_0) + h f_x(x_0, y_0) + k f_y(x_0, y_0) + h \epsilon_1(h, k) + k \epsilon_2(h, k)$$

where $\lim_{(h, k) \rightarrow (0, 0)} \epsilon_1(h, k) = \lim_{(h, k) \rightarrow (0, 0)} \epsilon_2(h, k) = 0$

Equivalently, let $\nabla f = (f_x, f_y)$, $\bar{P} = (x_0, y_0)$, $\bar{e} = (h, k)$, then

$$f(\bar{P} + \bar{e}) = f(\bar{P}) + \nabla f(\bar{P}) \cdot \bar{e} + \epsilon(\bar{e}) \cdot \bar{e}$$

where $\lim_{\bar{e} \rightarrow \bar{0}} \epsilon(\bar{e}) = 0$

It is not true that if $f_x(x_0, y_0)$ & $f_y(x_0, y_0)$ exist then $f(x, y)$ is differentiable at (x_0, y_0)

Eg $f(x, y) = -10 \sqrt{|xy|}$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-10\sqrt{0} - (-10\sqrt{0})}{h} = 0$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0$$

But if $\bar{e} = (h, h)$ $\bar{P} = (0, 0)$ then

$$f(h, h) = f(0, 0) + \cancel{\nabla f \cdot (h, h)} + \epsilon_1(h, h) \cdot h + \epsilon_2(h, h) \cdot h$$

$$-10\sqrt{|h^2|} = \epsilon \cdot h$$

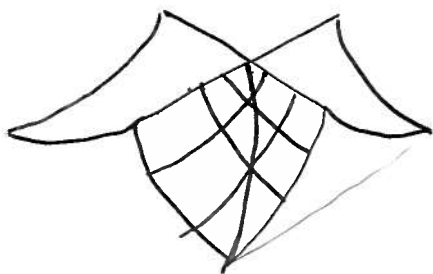
where $\epsilon = \epsilon_1 + \epsilon_2$

$$\epsilon = \frac{-10|h|}{h}$$

$$\lim_{h \rightarrow 0^+} \epsilon = -10$$

$$\lim_{h \rightarrow 0^-} \epsilon = 10$$

Not 0! Not differentiable!



Thm If $f_x(x, y)$ & $f_y(x, y)$ exist and are continuous on a disk containing (x_0, y_0) , then $f(x, y)$ is differentiable at (x_0, y_0) .

Ex $f(x, y) = -10 \sqrt{|x, y|}$

$$f_x(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(h, y_0) - f(y_0)}{h} = \lim_{h \rightarrow 0} \frac{-10 \sqrt{|h, y_0|}}{h}$$

$$= \lim_{h \rightarrow 0} -10 \sqrt{|y_0|} \cdot \frac{\sqrt{|h|}}{h} \quad \text{DNE if } y_0 \neq 0 !$$

Ex $f(x, y) = \sin(xy) \quad \left. \begin{array}{l} f_x = y \cos(xy) \\ f_y = x \cos(xy) \end{array} \right\} \begin{array}{l} \text{continuous} \\ \text{on } \mathbb{R}^2 \end{array}$

So $f(x, y)$ is differentiable on $\mathbb{R}^2$

Ex $f(x, y) = e^{x/y} \quad \left. \begin{array}{l} f_x = \frac{1}{y} e^{x/y} \\ f_y = -\frac{x}{y^2} e^{x/y} \end{array} \right\} \begin{array}{l} \text{continuous on} \\ \mathbb{R}^2 - \{y=0\} \end{array}$

So $f(x, y)$ is differentiable on $\mathbb{R}^2 - \{y=0\}$

Thm If $f(x, y)$ is differentiable at (x_0, y_0) then it is continuous at (x_0, y_0)

Pf We know $f(x_0+h, y_0+k) - f(x_0, y_0) = f_x \cdot h + f_y \cdot k + \epsilon_1(h, k) \cdot h + \epsilon_2(h, k) \cdot k$

where $\lim_{(h, k) \rightarrow (0, 0)} \epsilon_i = 0$

$$\begin{aligned} \text{Then } \lim_{(h, k) \rightarrow (0, 0)} [f(x_0+h, y_0+k) - f(x_0, y_0)] &= \\ &= \lim_{(h, k) \rightarrow (0, 0)} \left[\underbrace{f_x}_{\downarrow 0} \cdot h + \underbrace{f_y}_{\downarrow 0} \cdot k + \underbrace{\epsilon_1}_{\downarrow 0} \cdot h + \underbrace{\epsilon_2}_{\downarrow 0} \cdot k \right] = 0 \quad \checkmark \end{aligned}$$

Thm $\nabla f = (f_x, f_y)$ $\nabla(f+g) = ((f+g)_x, (f+g)_y) = (f_x + g_x, f_y + g_y) = \nabla f + \nabla g$

$$\nabla(\alpha f) = ((\alpha f)_x, (\alpha f)_y) = (\alpha f_x, \alpha f_y) = \alpha(f_x, f_y) = \alpha \nabla f$$

$$\begin{aligned} \nabla(f \circ g) &= ((f \circ g)_x, (f \circ g)_y) = (f_x \circ g + f_y \circ g_x, f_x \circ g + f_y \circ g_y) = g(f_x, f_y) + f(g_x, g_y) \\ &= g \nabla f + f \nabla g. \end{aligned}$$

Thm Let $f = f(x, y, z)$. The tangent plane to $f(x, y, z) = c$ at a point (x_0, y_0, z_0) is given by

$$\nabla f(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

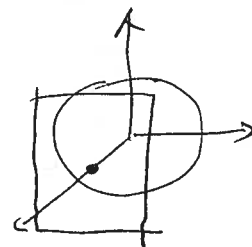
Ex

Ex Find tangent plane to

$$f(x, y, z) = x^2 + y^2 + z^2 = 1 \quad \text{at } (1, 0, 0) \text{ \& } (\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}})$$

$$\nabla f = (2x, 2y, 2z) \quad \nabla f(1, 0, 0) = (2, 0, 0)$$

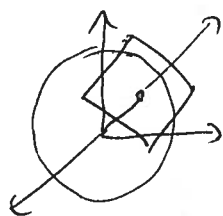
$$0 = (2, 0, 0) \cdot (x-1, y-0, z-0) = 2x-2$$



$$\nabla f(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}) = (1, 1, \sqrt{2})$$

$$0 = (1, 1, \sqrt{2}) \cdot (x-\frac{1}{2}, y-\frac{1}{2}, z-\frac{1}{\sqrt{2}}) = x-\frac{1}{2} + y-\frac{1}{2} + \sqrt{2}z-1$$

$$= x+y+\sqrt{2}z-2$$



Ex Find tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$

$$\Rightarrow x^2 + y^2 - z = 0 \quad \nabla f = (2x, 2y, -1) \quad \nabla f(1, 1, 2) = (2, 2, -1)$$

$$0 = (2, 2, -1) \cdot (x-1, y-1, z-2) = 2x-2+2y-2-z+2 = 2x+2y-z-2$$

Ex Find tangent plane to $z = \sin(x-y)$ at $(0, 0, 0)$

$$f(x, y, z) = \sin(x-y) - z \quad \nabla f = (\cos(x-y), -\cos(x-y), -1)$$

$$\nabla f(0, 0, 0) = (1, -1, -1)$$

$$0 = (1, -1, -1) \cdot (x, y, z) = x-y-z$$

$z = x-y$ is 1st order approximation of
 $z = \sin(x-y)$
 near $(0, 0)$.

Exercises.

1) Compute ∇f where $f(x, y) = xe^{xy}$, $\frac{x^2y}{(x+y)}$, $\sin^3(x^2y)$

2) Find the equation of the tangent plane at P .

a) $f(x, y) = x^2y - xy^2$ $P = (-2, 3)$

b) $f(x, y) = \frac{x^2}{y}$ $P = (2, -1)$

c) $f(x, y, z) = xyz + x^2$ at $P = (1, 2, -1)$

3) Find the points (x, y) at which the tangent plane to $z = x^2 - 6x + 2y^2 - 10y + 2xy$ is horizontal.